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Senior Software Engineer | Tech Lead | Production LLM Systems & Agentic Engineering
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SUMMARY

Senior engineer and Tech Lead with 10+ years delivering enterprise systems in banking, insurance, public sector and ECM. Currently lead a team of 5 and the AI architecture of a production document-processing platform with self-hosted vision-language model inference on a private AWS GPU node (vLLM, NVIDIA A10G, private VPC). I ship LLM systems under real constraints — cost, latency, deterministic validation and human-in-the-loop authority — and work AI-native every day with Claude Code and OpenAI Codex: skills, harness and agentic-loop engineering (CLAUDE.md / AGENTS.md, hooks, custom skills library).

CORE STACK

AI / LLM engineering: Self-hosted VLM inference (Ollama, vLLM, NVIDIA L4/A10G on AWS EC2 g6.xlarge/g5.xlarge, private VPC), inference-avoidance gating, per-invocation telemetry, latency & cost measurement, deterministic validation, explicit abstention, human review, Claude API

Agentic engineering: Claude Code and OpenAI Codex (daily), skills engineering, harness engineering (CLAUDE.md / AGENTS.md, hooks, permissions), agentic-loop design (plan -> act -> verify -> iterate), verification gates and evaluation baselines

Languages: C#, Python, Java, JavaScript, T-SQL, VB.NET

Backend: .NET 10 / .NET Core / ASP.NET MVC, Dapper, Entity Framework, ADO.NET, LINQ, REST APIs, SignalR, Spring

Data & Cloud: SQL Server, AWS (EC2 GPU, VPC), Git, Linux

ECM & Automation: OnBase Workflow (Hyland certified), RPA: UiPath, Jidoka, Appian

Frontend & Process: jQuery, Kendo UI, Bootstrap, Figma · Scrum (certified), DevOps, Lean

EXPERIENCE

Development Supervisor & AI Architect · **SEA Servicios Múltiples** Jul 2025 – Present · *Remote*

- Lead a team of 5 engineers and the AI architecture of **DocumentPipelineAI**, a production intelligent document processing (IDP) platform on .NET 10, Dapper and SQL Server, with self-hosted vision-language model inference on a private AWS GPU node (NVIDIA A10G, EC2 g5.xlarge) served through vLLM inside a private VPC.
- Designed the system as AI-first but not AI-authoritative: the model handles perception; deterministic validation, explicit abstention and human review keep authority over published data.
- Reduced per-page latency ~22% (≈59 s -> ≈46 s) with measurement-driven optimization; instrumented per-invocation telemetry and corrected the optimizable-fraction estimate from 42% to 76% by measuring instead of assuming.
- Built inference-avoidance gates that resolve ~40% of requests without invoking the model, cutting GPU load and cost per document.
- Diagnosed an HttpClient.Timeout defect in the inference transport and locked the fix with a regression test verified by mutation testing.
- Defined a per-digit accuracy criterion for 50-digit invoicing keys and an optimization roadmap with explicit go/no-go criteria per option (evaluated, not deployed: speculative decoding, LoRA/QLoRA fine-tuning, roofline modeling).

Full Stack Engineer · **Scotiabank** Nov 2022 – Jun 2025 · *Costa Rica*

- Led the migration of banking systems from Java 8 / WebSphere to Java 17 / TomEE, improving scalability and performance.
- Developed and supported the Quickpass system, ensuring functionality and reliability.

- Built and maintained automated workflows integrating MVC architecture with jQuery, Kendo UI and SQL Server.

Tech Lead & Developer · **SEA Servicios Múltiples** Aug 2017 – Oct 2022 · *Costa Rica*

- Implemented OnBase workflows for credit processing, optimizing database performance and user interfaces.
- Built a public-security operations control system with real-time communication (SignalR) and secure digital signing.
- Built an SMS campaign management and automation system with real-time delivery tracking.

Full Stack Developer · **RegionalSoft** Jun 2013 – Jul 2017 · *Costa Rica*

- Designed and developed a responsive web application with SEO optimization and dynamic data visualization.
- Migrated and modernized a desktop application to Windows 10 and enhanced FileMaster, improving data reliability and user experience.

PERSONAL PROJECTS

Lalia · **personalized language-tutor agent on Hermes Agent** 2026

- Adaptive language tutor (English first) built on Hermes Agent and delivered over Telegram: spaced-repetition cards, XP and streaks, voice-note practice with local speech-to-text (faster-whisper, word-level timestamps) and text-to-speech (piper), and a per-learner error fingerprint that tracks recurring mistakes and retrieval stages.
- Methodology roadmap grounded in second-language-acquisition research rather than generic chat: Nation's four strands and 4/3/2 fluency drills, Refold-style i+1 sentence mining, self-repair prompts (Lyster & Ranta) with input flooding, dictogloss, and automatic CAF metrics (speech rate, pause ratio, errors per 100 words) computed from word-level timestamps.
- Status: working prototype in personal use; I am its first user for my own English.

AGENTIC ENGINEERING: SKILLS, HARNESS & LOOP

- **Skills engineering:** authored a personal library of modular, reusable skills (spec elicitation with epistemic tags, detailed design, production code-quality gates, test strategy, secure content processing) with explicit hand-offs between them, so agents follow a spec -> design -> code -> test pipeline instead of ad-hoc prompting.
- **Harness engineering:** design and audit agent harnesses for Claude Code and Codex (CLAUDE.md / AGENTS.md, hooks, permissions); classify failure modes — context, constraint, verification, planning — and map each one to a harness component; ship changes as mini-ADRs with confirmation gates and an evaluation baseline.
- **Agentic-loop engineering:** design the outer loop (plan -> act -> verify -> iterate) with deterministic verification hooks, explicit stop and escalation criteria, measurement before optimization, and a clear line between what is deployed and what is only evaluated.

CERTIFICATIONS & EDUCATION

- **OnBase Certified Workflow Engineer for Partners** — Hyland, 2024
- Scrum Master — MAPO, 2018 · Microsoft Certified Professional — 2014
- Creative UX/UI: User Experience & Interface Design (105 h) — Universidad Véritas, 2021
- Embedded Systems & IoT technical program — Universidad Fidélitas, 2024
- Mobile Development (Xamarin) — Cenfotec, 2020
- B.Eng. Systems Engineering — UAM (in progress)

LANGUAGES

Spanish — native · English — B1 (technical reading and writing; async written collaboration) · Portuguese — A2